

Postpartum Medicaid Access Barrier Index

Project Proposal for Index Development, Validation, and Policy-Facing Dashboard Research

Prepared for the Postpartum Medicaid Access Barriers Power BI portfolio project

Overview

This project proposes a county-level Postpartum Medicaid Access Barrier Index designed to identify places where postpartum Medicaid populations may face overlapping administrative, transportation, digital, linguistic, rural, and maternity-care access barriers. The index is intended as a transparent screening measure for policy analysis and dashboard reporting, not as a causal estimate of postpartum outcomes.

Main contribution: a reproducible, interpretable county-level indicator that highlights where limited Medicaid office availability, maternity care access constraints, and community-level access barriers overlap.

Primary Research Question

Can a county-level composite index identify geographic areas where postpartum Medicaid populations may face elevated administrative and service access barriers after delivery?

Secondary questions include:

- Which counties score highest on the index?
- Are high-barrier counties concentrated in particular regions or rurality categories?
- How much of the index is driven by administrative access versus clinical maternity access?
- Does the index identify counties that would be missed by looking only at Medicaid office availability or obstetric care access alone?

Conceptual Framework

The index is organized around two linked domains. The first is administrative access: whether postpartum Medicaid beneficiaries may face barriers to maintaining coverage, resolving paperwork, navigating eligibility systems, or getting in-person support. The second is clinical maternity access context: whether counties have hospital-based obstetric care infrastructure that may support pregnancy and postpartum care continuity.

The framework treats postpartum Medicaid access as a systems-navigation problem shaped by geography, transportation, digital access, language access, disability-related support needs, poverty, reproductive-age population context, and rurality.

Planned Index Design

Component	Points	Interpretation
No Medicaid office in county	2	Administrative/enrollment support access measure
No hospital-based obstetric care	2	Clinical maternity access context
Top quartile poverty rate	1	County-level economic need
Top quartile no-vehicle household rate	1	Transportation barrier
Top quartile no-internet subscription rate	1	Digital access barrier
Top quartile limited English rate	1	Language access context
Top quartile disability rate	1	Access-support context measure
Top quartile female population ages 15-44 rate	1	Reproductive-age population proxy
Nonmetro county	1	Rural access context

Planned score range: 0-11 points. Planned concern levels: 0-2 lower concern, 3-5 moderate concern, 6-8 high concern, and 9-11 highest concern.

Validation Strategy

Face Validity

Examine whether highest-scoring counties align with expected access-barrier profiles, such as no Medicaid office, no obstetric care, high poverty, limited transportation, low internet access, and nonmetro status.

Construct Validity

Compare the index with related but distinct measures such as rurality, Social Vulnerability Index if added later, maternity care desert classifications, broadband access, vehicle access, poverty, and Medicaid office availability.

Known-Groups Validity

Test whether scores differ across groups expected to vary, including metro versus nonmetro counties, counties with versus without hospital-based obstetric care, and counties with versus without Medicaid offices.

Sensitivity Testing

Recalculate the index under alternative specifications, such as equal weights, no Medicaid office component, no obstetric care component, tertile thresholds, continuous z-score indicators, and versions excluding poverty.

Geographic Validation

Use maps, regional summaries, state-level comparisons, and spatial clustering diagnostics to determine whether high-barrier counties form interpretable geographic patterns.

External Outcome Validation

If appropriate data are available, test whether higher index scores are associated with related outcomes or context measures such as postpartum visit attendance, Medicaid churn, maternity care desert status, provider shortage areas, or maternal health access indicators.

Recommended Statistical Analyses

- Descriptive statistics for all index components and score distribution.
- County and population-weighted summaries where appropriate.
- Cross-tabs by rurality, Medicaid office availability, and obstetric care status.
- Spearman correlations between the index and related access/context measures.
- Kruskal-Wallis or ANOVA comparisons across concern levels.
- Sensitivity analyses comparing alternative index specifications.
- Optional spatial autocorrelation or hot spot analysis to assess clustering.

Potential Tables and Figures

- Table 1. Index components, data sources, scoring rationale, and point values.
- Table 2. County characteristics by concern level.
- Figure 1. National county map of index score.
- Figure 2. Distribution of county index scores.
- Figure 3. Component contribution among highest-concern counties.
- Figure 4. Metro versus nonmetro score comparison.
- Figure 5. Sensitivity analysis of high-concern county classification.

Limitations

- The index does not identify individual postpartum Medicaid enrollees.
- Female population ages 15-44 is a reproductive-age population proxy, not a direct measure of postpartum need.
- Medicaid office availability does not capture online support, phone support, navigators, health systems, managed care plans, or community-based organizations.
- County-level measures may hide within-county variation.
- Hospital-based obstetric care status is not the same as comprehensive postpartum clinical access.
- The index is not causal and should be interpreted as a screening tool.
- Weights are theory-informed and should be tested through sensitivity analyses.
- State Medicaid administrative structures differ, so county office availability may not mean the same thing in every state.

Recommended Next Step

The next step is to develop a formal index validation plan that specifies hypotheses, data sources, validation tests, sensitivity analyses, and planned tables and figures. This document can serve as the foundation for both the research paper methods section and the Power BI dashboard methodology page.